COL8628/COL828 Assignment 1

Transformer Models for Implant Classification

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September 2025

Abstract

This report investigates the application of Vision Transformer (ViT) models and prompt learning techniques for medical implant classification across two datasets: Orthonet (orthopedic knee implants) and Pacemakers (cardiac devices). We evaluate multiple pretraining paradigms including ImageNet-21k, CLIP, and DINOv2, assess zero-shot classification capabilities, and explore prompt learning methods (CoOp, CoCoOp, MaPLe) in both flat and hierarchical settings. Our results demonstrate that supervised pretraining on ImageNet-21k achieves superior performance on Orthonet, while MaPLe with hierarchical prompting significantly outperforms other methods on the fine-grained Pacemaker dataset, achieving 71.56% top-1 accuracy.

1 Introduction

Medical image analysis, particularly implant classification from X-ray images, presents unique challenges due to domain specificity, limited training data, and the need for fine-grained discrimination. This assignment explores how Vision Transformers pretrained under different paradigms transfer to medical imaging tasks and investigates whether prompt learning can bridge the domain gap without extensive fine-tuning.

We conduct experiments on two datasets with contrasting characteristics: Orthonet, containing orthopedic knee implants, and Pacemakers, featuring a hierarchical taxonomy of 45 cardiac device models across 5 manufacturers. Due to computational constraints, hyperparameter optimization was limited, and we focus on comparing architectural choices and learning paradigms. Training plots can be found here.

2 Task 1: Orthonet Dataset

The Orthonet dataset consists of X-ray images of orthopedic knee implants, representing a relatively coarse-grained classification problem compared to the Pacemaker dataset.

2.1 Task 1.1: Fine-tuning Pretrained ViT Variants

We fine-tuned three ViT variants initialized with different pretrained weights: ImageNet-21k (supervised), CLIP (vision-language), and DINOv2 (self-supervised).

Training Configuration: All models were trained for 15 epochs using AdamW optimizer with learning rate 2e-5, cosine annealing learning rate scheduler with standard parameters, and default weight decay.

Table 1: Performance comparison of fine-tuned ViT variants on Orthonet

Model	Test Accuracy (%)	F1-Score	AUC-ROC
ImageNet-21k	**85.56**	**0.8479**	**0.9927**
CLIP	87.22	0.8629	0.9912
DINOv2	82.78	0.8112	0.9867

Key Observations:

- **CLIP achieves highest accuracy (87.22%)**, suggesting that vision-language pretraining provides representations that transfer well to medical imaging despite being trained on natural images with text supervision.
- **ImageNet-21k demonstrates superior AUC-ROC (0.9927)**, indicating better calibrated confidence scores and ranking quality, which is crucial for medical applications where uncertainty quantification matters.
- **DINOv2 underperforms** (82.78% accuracy), which is surprising given its strong performance on natural images. This suggests that self-supervised learning objectives optimized for semantic understanding may not capture the fine textural and structural details critical for implant classification.
- The relatively small performance gap (4.44% between CLIP and DINOv2) indicates that with sufficient fine-tuning, different pretraining paradigms can converge to similar performance levels.

2.2 Task 1.2: Zero-Shot Classification with CLIP

Zero-shot CLIP classification was evaluated using handcrafted prompts of the form *"an X-ray of a {class} implant"*.

Table 2: Zero-shot CLIP performance on Orthonet

Metric	Value
Top-1 Accuracy	0.1667
F1-Score (Macro)	0.0911
F1-Score (Weighted)	0.1004
AUC-ROC	0.7543

Analysis:

The zero-shot performance is surprisingly poor (16.67% accuracy), revealing a significant domain gap between CLIP's natural image-text pretraining and medical X-ray classification. This 70.55% performance drop compared to fine-tuned CLIP (87.22%) highlights several critical insights:

- **Medical terminology mismatch**: CLIP's text encoder was not exposed to medical imaging vocabulary during pretraining, leading to weak image-text alignment for specialized implant types.
- **Visual distribution shift**: X-ray images have fundamentally different characteristics (grayscale, specific anatomical structures, metallic implants) compared to natural RGB images in CLIP's training data.

- **Fine-grained discrimination**: Distinguishing between similar implant models requires subtle feature detection that simple prompt templates cannot capture without task-specific adaptation.

Despite poor accuracy, the AUC-ROC of 0.7543 suggests CLIP retains some ranking ability, indicating partial semantic understanding.

2.3 Task 1.3: Prompt Learning Methods

To bridge the domain gap without full fine-tuning, we evaluated three prompt learning approaches: CoOp, CoCoOp, and MaPLe. All methods kept the CLIP backbone frozen and only learned prompt tokens.

Table 3: Prompt learning results on Orthonet with hyperparameters

Method	Hyperparameters	Acc.	F1 (M)	F1 (W)	AUC
CoOp	50ep, lr=2e-3, AdamW	0.3500	0.2906	0.2906	0.8319
CoOp	50ep, lr=1e-3, SGD	0.4278	0.3647	0.3647	0.8723
CoCoOp	50ep, lr=2e-3, AdamW	0.3778	0.3197	0.3197	0.8321
CoCoOp	50ep, lr=1e-3, AdamW	0.2278	0.1505	0.1505	0.7921
MaPLe	60ep, lr=1e-3, SGD, wd=5e-4, warmup=1ep, N_CTX=2	**0.6667**	**0.6517**	**0.6517**	**0.9449**

Implementation Details:

- **CoOp & CoCoOp**: Standard implementations with 50 training epochs and cosine annealing scheduler. Both AdamW and SGD optimizers were tested with learning rates 2e-3 and 1e-3.
- **MaPLe**: Configured with N_CTX=2 (2 context tokens), PROMPT_DEPTH=9 (prompts at 9 transformer layers), fp32 precision, SGD optimizer (lr=1e-3, weight decay=5e-4), 60 epochs with cosine annealing, 1 warmup epoch (warmup lr=1e-5).

Key Findings:

- **MaPLe achieves the best prompt learning performance** (66.67% accuracy), dramatically outperforming CoOp (42.78%) and CoCoOp (37.78%) by 23.89% and 28.89% respectively. This represents a 50% improvement over zero-shot CLIP (16.67%), demonstrating the effectiveness of multi-modal, multi-layer prompting.
- **Multi-layer prompt distribution is critical**: MaPLe's superior performance validates its architectural innovation of distributing learnable prompts across 9 transformer layers in both vision and text encoders, enabling deeper adaptation to the medical imaging domain compared to single-layer text-only prompts in CoOp/CoCoOp.
- **Hyperparameter choices significantly impact performance**:
  - **Optimizer selection**: CoOp with SGD (lr=1e-3) outperforms AdamW (lr=2e-3) by 7.78%, suggesting momentum-based optimization provides better convergence for simpler prompt learning architectures.
  - **Learning rate sensitivity**: CoCoOp's performance drops dramatically from 37.78% (lr=2e-3) to 22.78% (lr=1e-3), indicating high sensitivity to learning rate that may require task-specific tuning.

  - **Training duration**: MaPLe required 60 epochs vs 50 for CoOp/CoCoOp, suggesting deeper architectures benefit from longer training.
  - **Weight decay**: MaPLe uses explicit weight decay (5e-4) for regularization, which may contribute to better generalization.

- **MaPLe's AUC-ROC (0.9449) approaches fine-tuned models** (0.9912-0.9927), indicating excellent confidence calibration and ranking ability, which is crucial for medical applications where uncertainty quantification matters.

- **CoCoOp's input-conditioned prompts do not improve performance**, contrary to expectations. The best CoCoOp variant (37.78%) underperforms both CoOp (42.78%) and MaPLe (66.67%), potentially because instance-specific adaptation requires more training data or introduces optimization challenges with frozen backbones.

- **Prompt learning still trails full fine-tuning**: Despite MaPLe's strong performance (66.67%), it remains 20.55% below fine-tuned CLIP (87.22%), indicating that freezing the CLIP backbone imposes a fundamental performance ceiling, though the gap is substantially narrower than with simpler prompt learning methods.

3 Task 2: Pacemaker Dataset

The Pacemaker dataset presents a more challenging fine-grained classification problem with 45 implant classes organized hierarchically across 5 manufacturers.

3.1 Task 2.1: Fine-tuning Pretrained ViT Variants

Training Configuration: All models were trained for 15 epochs using AdamW optimizer with learning rate 2e-5, cosine annealing learning rate scheduler with standard parameters, and default weight decay.

Table 4: Fine-tuned ViT performance on Pacemaker dataset

Model	Test Accuracy (%)	F1-Score	AUC-ROC
ImageNet-21k	**93.78**	**0.9371**	**0.9990**
CLIP	63.11	0.6335	0.9771
DINOv2	88.89	0.8882	0.9970

Critical Observations:

- **ImageNet-21k dramatically outperforms other methods** (93.78% vs 88.89% for DINOv2 and 63.11% for CLIP), suggesting that supervised pretraining on diverse object categories provides superior transferable features for fine-grained discrimination.

- **CLIP's performance collapse** (30.67% drop from ImageNet-21k) is striking and indicates that vision-language alignment objectives may not prioritize the discriminative visual features necessary for distinguishing highly similar implant models. This contrasts with Task 1 where CLIP performed best, suggesting dataset-specific transferability patterns.

- **DINOv2 recovers strongly** (88.89%), closing the gap with supervised pretraining compared to Task 1. This suggests self-supervised features excel when fine-grained visual discrimination is paramount.

- **Near-perfect AUC-ROC scores** (0.997-0.999) indicate excellent confidence calibration across all methods, despite accuracy differences.

3.2 Task 2.2: Zero-Shot CLIP Classification

Table 5: Zero-shot CLIP on Pacemaker dataset

Metric	Value
Top-1 Accuracy	0.0133
F1-Score (Macro)	0.0037
F1-Score (Weighted)	0.0037
AUC-ROC	0.4764

Analysis:

Zero-shot performance is catastrophically poor (1.33% accuracy, near random chance for 45 classes at 2.22%). This represents complete failure and is worse than Orthonet's zero-shot performance, revealing:

- **Fine-grained discrimination failure**: With 45 visually similar classes, handcrafted prompts cannot encode sufficient discriminative information.
- **Hierarchical complexity**: The manufacturer-series-model taxonomy requires structured reasoning that simple prompt templates cannot capture.
- **AUC-ROC of 0.4764** (below random 0.5) suggests anti-correlation, indicating the text encoder's semantic space is fundamentally misaligned with the visual features needed for pacemaker classification.

3.3 Task 2.3: Hierarchical Prompt Learning

Given the hierarchical structure of the Pacemaker dataset, we evaluated flat and hierarchical prompt learning strategies.

3.3.1 Flat Prompt Learning

All flat prompt learning methods trained for 50 epochs with cosine annealing scheduler and 1 epoch warmup (warmup lr=1e-5).

Table 6: Flat prompt learning on 45-class Pacemaker classification

Method	Hyperparameters	Top-1	Top-3	F1 (M)	F1 (W)	AUC
CoOp	lr=1e-3, AdamW	0.4667	0.6889	0.4417	0.4417	0.9338
CoCoOp	lr=1e-3, AdamW	0.3733	0.6667	0.3539	0.3539	0.9259
MaPLe	lr=2.5e-4, AdamW, wd=5e-4, N_CTX=10, DEPTH=12	**0.7511**	**0.9289**	**0.7488**	**0.7488**	**0.9876**

MaPLe demonstrates exceptional performance (75.11% accuracy), dramatically outperforming CoOp (46.67%) and CoCoOp (37.33%). This 28.44% improvement validates MaPLe's multi-modal, multi-layer prompt distribution strategy with deeper prompt injection (PROMPT_DEPTH=12) and more context tokens (N_CTX=10), which enables:

- Stronger image-text alignment through prompts in both encoders
- Hierarchical feature adaptation across transformer layers
- Better fine-grained discrimination capability with increased prompt capacity

Hyperparameter Configuration Impact:

- MaPLe uses a lower learning rate (2.5e-4) compared to CoOp/CoCoOp (1e-3), suggesting that deeper prompt architectures require more conservative optimization.
- The increased N_CTX (10 vs typical 2-4) and PROMPT_DEPTH (12 layers) provide substantially more learnable parameters, enabling better adaptation to the fine-grained 45-class problem.

3.3.2 Hierarchical Prompt Learning

We evaluated a two-stage approach: Stage 1 learns manufacturer-level (5-class) prompts, which initialize Stage 2's fine-grained (45-class) prompts. All methods use cosine annealing scheduler with 1 epoch warmup.

Training Configuration:

- **CoOp**: Stage 1: lr=1e-3, 50 epochs, weight decay=1e-4; Stage 2: lr=5e-4, 100 epochs, weight decay=1e-4, AdamW
- **CoCoOp**: Stage 1: lr=5e-4, 50 epochs, weight decay=1e-4; Stage 2: lr=2.5e-4, 100 epochs, weight decay=1e-4, AdamW
- **MaPLe**: Stage 1 and Stage 2 hyperparameters matching the flat configuration (specific values not provided, but consistent with hierarchical initialization approach)

Stage 1: Manufacturer-Level Classification

Table 7: Stage 1 manufacturer-level (5-class) results

Method	Top-1	Top-3	F1 (Macro)	F1 (Weighted)	AUC-ROC
CoOp	0.6578	0.9378	0.5909	0.6467	0.8876
CoCoOp	0.6578	0.9289	0.6137	0.6542	0.8986
MaPLe	**0.8489**	**0.9822**	**0.8414**	**0.8479**	**0.9720**

At the manufacturer level, MaPLe achieves 84.89% accuracy, substantially outperforming CoOp and CoCoOp (both at 65.78%), demonstrating its superiority even for coarser-grained classification.

Stage 2: Fine-Grained Classification with Hierarchical Initialization

Table 8: Stage 2 fine-grained (45-class) results with hierarchical initialization

Method	Top-1	Top-3	F1 (Macro)	F1 (Weighted)	AUC-ROC
CoOp	0.4267	0.6533	0.4197	0.4197	0.9265
CoCoOp	0.4133	0.6489	0.4026	0.4026	0.8947
MaPLe	**0.7156**	**0.8978**	**0.7126**	**0.7126**	**0.9875**

3.3.3 Flat vs. Hierarchical Comparison

Table 9: Impact of hierarchical initialization (Top-1 Accuracy)

Method	Flat	Hierarchical	Difference
CoOp	0.4667	0.4267	-0.0400 (-8.57%)
CoCoOp	0.3733	0.4133	+0.0400 (+10.71%)
MaPLe	0.7511	0.7156	-0.0355 (-4.73%)

Surprising Results:

- **Hierarchical initialization does not consistently improve performance**. MaPLe and CoOp perform slightly worse with hierarchical prompts, while CoCoOp shows modest improvement.
- **Learning rate scheduling in hierarchical training**: Both CoOp and CoCoOp use staged learning rate reduction (Stage 1: higher lr, fewer epochs; Stage 2: lower lr, more epochs - 100 epochs vs 50). Specifically, CoOp uses lr=1e-3→5e-4 and CoCoOp uses lr=5e-4→2.5e-4, following the principle that fine-grained refinement requires more conservative optimization.
- **MaPLe's flat performance (75.11%) remains best overall**, suggesting that with sufficient model capacity (N_CTX=10, PROMPT_DEPTH=12) and multi-layer prompting, direct 45-class learning may be more effective than hierarchical decomposition.
- **CoCoOp benefits from hierarchical structure** (+10.71%), potentially because instance-conditioned prompts can better exploit manufacturer-level features as a scaffold for fine-grained discrimination. The staged learning rate reduction may also help CoCoOp's meta-network converge more stably.
- The performance decrease for MaPLe (-4.73%) and CoOp (-8.57%) may indicate that hierarchical initialization introduces constraints that limit the model's ability to discover optimal prompt configurations directly from the 45-class objective. The additional 50 epochs of Stage 2 training (100 vs 50 in flat) do not compensate for suboptimal initialization.

4 Cross-Task Analysis

4.1 Pretraining Paradigm Effectiveness

Table 10: Pretraining paradigm performance across tasks

Method	Orthonet (Accuracy)	Pacemaker (Accuracy)
ImageNet-21k	85.56%	**93.78%**
CLIP	**87.22%**	63.11%
DINOv2	82.78%	88.89%

The contrasting performance patterns reveal important insights:

- **CLIP excels on Orthonet but collapses on Pacemaker**, suggesting its vision-language representations work well for moderate-difficulty medical classification but fail when fine-grained discrimination is required.

- **ImageNet-21k supervised pretraining is most robust**, achieving strong performance on both datasets (85.56% and 93.78%), validating the effectiveness of diverse object-level features for transfer learning.
- **DINOv2 shows consistent mid-tier performance**, performing better on fine-grained Pacemaker (88.89%) than coarse Orthonet (82.78%), suggesting self-supervised features capture discriminative patterns that scale with task complexity.

4.2 Prompt Learning Effectiveness

- **MaPLe consistently dominates prompt learning across both datasets**, achieving 66.67% on Orthonet and 75.11% (flat) / 71.56% (hierarchical) on Pacemaker. This demonstrates that multi-modal, multi-layer prompting is essential for effective medical image classification without backbone fine-tuning.
- **CoOp and CoCoOp struggle with both datasets**, achieving only 42.78% (Orthonet) and 46.67% (Pacemaker flat), indicating that single-layer text-only prompts have limited capacity for medical domain adaptation. MaPLe's 23.89% improvement over CoOp on Orthonet validates the importance of vision-side prompting and multi-layer distribution.
- **Task complexity affects the prompt learning gap**: On Orthonet, MaPLe achieves 76.5% of fine-tuned CLIP performance (66.67% vs 87.22%), while on Pacemaker it reaches 80.1% of ImageNet-21k performance (75.11% vs 93.78%), suggesting prompt learning is increasingly competitive as task complexity increases.
- **Zero-shot CLIP is inadequate for medical imaging** (16.67% Orthonet, 1.33% Pacemaker), reinforcing the necessity of task-specific adaptation. However, MaPLe bridges 80.2% of the zero-shot to fine-tuned gap on Orthonet, demonstrating effective parameter-efficient adaptation.

5 Discussion

5.1 Implications for Medical Imaging

Our experiments reveal critical considerations for applying vision transformers to medical imaging:

1. **Supervised pretraining remains gold standard**: Despite advances in self-supervised and vision-language models, ImageNet-21k supervised pretraining provides the most reliable transfer learning, especially for fine-grained medical tasks.
2. **Domain gap is severe for vision-language models**: CLIP's poor zero-shot and fine-tuned performance on Pacemaker highlights fundamental incompatibility between natural image-text pretraining and specialized medical imaging.
3. **Prompt learning shows promise but limitations**: While MaPLe achieves respectable 75.11% accuracy without fine-tuning the backbone, it still underperforms full fine-tuning (93.78%), suggesting prompt learning is a parameter-efficient alternative but not a performance-optimal solution.
4. **Hierarchical structure is underutilized**: Our hierarchical prompt learning approach did not yield consistent improvements, suggesting that more sophisticated methods for exploiting taxonomic structure (e.g., hierarchical loss functions, conditional computation) may be necessary.

5. **Hyperparameter sensitivity varies by architecture**: Simple prompt methods (CoOp, CoCoOp) show high sensitivity to optimizer choice and learning rate, while deeper architectures (MaPLe) demonstrate more robust performance across configurations but require careful tuning of prompt depth and context length.

5.2 Hyperparameter Insights and Recommendations

Our experiments across multiple prompt learning configurations reveal several practical guidelines:

- **Architecture-specific optimization**: Shallow prompt methods (CoOp) benefit from SGD with momentum, while deeper architectures (MaPLe) achieve better results with AdamW and conservative learning rates.
- **Learning rate scheduling**: All successful configurations employed cosine annealing, which appears critical for prompt learning convergence. Hierarchical approaches benefit from staged learning rate reduction (higher lr for coarse classes, lower lr for fine-grained refinement).
- **Prompt capacity scaling**: For fine-grained tasks (45 classes), increasing both N_CTX (2→10) and PROMPT_DEPTH (9→12) substantially improves performance, suggesting that prompt capacity should scale with task complexity.
- **Training duration**: Deeper prompt architectures (MaPLe: 60 epochs) require longer training than shallow methods (CoOp/CoCoOp: 50 epochs), but hierarchical approaches need even more total epochs (Stage 1: 50 + Stage 2: 100 = 150 total).
- **Weight decay regularization**: Explicit weight decay (5e-4 for MaPLe, 1e-4 for hierarchical methods) appears important for preventing overfitting in prompt learning, especially with larger prompt capacities.
- **Warmup is beneficial**: All successful configurations included 1 epoch of learning rate warmup (warmup lr=1e-5), suggesting that gradual optimization startup helps prompt learning stability.

5.3 Limitations and Future Work

Due to computational constraints, this work has several limitations:

- **Limited hyperparameter tuning**: While we explored multiple learning rates (2e-5 for fine-tuning; 1e-3, 2e-3, 2.5e-4, 5e-4 for prompt learning) and optimizers (AdamW, SGD), the search space was constrained. Systematic grid search or automated hyperparameter optimization could yield further improvements, particularly for prompt depth, context length, and weight decay values.
- **Fixed training schedules**: All fine-tuning experiments used 15 epochs, while prompt learning used 50-100 epochs. Optimal training duration may vary by dataset and method, and early stopping strategies were not explored.
- **No data augmentation analysis**: Medical imaging often benefits from domain-specific augmentations (e.g., rotation, contrast adjustment), which were not systematically explored. Different augmentation strategies may interact with pretraining paradigms.
- **Single model architecture**: All experiments used ViT; exploring other architectures (Swin Transformers, ConvNeXt) could reveal different transfer learning characteristics.

- **Hierarchical design space**: We tested only one hierarchical prompt learning approach with specific staged learning rates; alternative strategies (shared vs. separate prompts, different initialization schemes, varied epoch allocation) warrant investigation.
- **Limited prompt configuration exploration**: MaPLe's N_CTX and PROMPT_DEPTH were set based on initial experiments (N_CTX=2 for Orthonet, N_CTX=10 for Pacemaker; PROMPT_DEPTH=9 vs 12), but the optimal values may differ across tasks and datasets.

Future work should investigate:

1. Medical image-specific pretraining for CLIP-style models
2. Combining prompt learning with adapter modules for better efficiency-performance trade-offs
3. Explicit hierarchical modeling through structured prediction frameworks
4. Uncertainty quantification and calibration for clinical deployment
5. Automated hyperparameter optimization for prompt learning methods
6. Task-adaptive prompt capacity scaling strategies

6 Conclusion

This assignment comprehensively evaluated Vision Transformers and prompt learning techniques for medical implant classification. Key findings include:

1. ImageNet-21k supervised pretraining provides the most robust transfer learning across both coarse (Orthonet) and fine-grained (Pacemaker) medical imaging tasks, achieving 85.56% and 93.78% accuracy respectively with only 15 epochs of fine-tuning.
2. CLIP's performance is task-dependent: effective on Orthonet (87.22%) but collapses on fine-grained Pacemaker classification (63.11%), revealing limitations of vision-language pretraining for specialized medical domains requiring fine-grained discrimination.
3. MaPLe substantially outperforms other prompt learning methods, achieving 66.67% on Orthonet and 75.11% accuracy on 45-class Pacemaker classification without backbone fine-tuning. This demonstrates that multi-modal, multi-layer prompting with adequate capacity (N_CTX=10, PROMPT_DEPTH=12) is essential for medical domain adaptation.
4. Hyperparameter choices critically impact prompt learning: SGD benefits shallow methods (CoOp), while AdamW with conservative learning rates (2.5e-4) suits deeper architectures (MaPLe). Cosine annealing scheduling and learning rate warmup are essential across all methods.
5. Hierarchical prompt learning did not consistently improve performance over flat approaches, with MaPLe showing a 4.73% decrease (75.11%→71.56%) despite staged learning rate reduction and doubled training epochs. This suggests that direct end-to-end learning may be more effective than curriculum-based hierarchical decomposition for prompt optimization.
6. Zero-shot CLIP is inadequate for medical imaging (16.67% Orthonet, 1.33% Pacemaker), emphasizing the critical need for domain-specific adaptation. However, prompt learning methods can bridge a substantial portion of this gap without expensive full model fine-tuning.

These results underscore both the promise and limitations of transfer learning for medical imaging, highlighting the continued importance of supervised pretraining while opening avenues for more parameter-efficient adaptation methods. The significant performance impact of architectural choices (prompt depth, context length) and optimization hyperparameters (learning rate, optimizer, weight decay) suggests that careful configuration is essential for successful prompt learning deployment in medical imaging applications.

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